Mark Solms ~ Active Inference Insights 017 ~ Affect, Consciousness, Dreams

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hello everybody and welcome back to active inference insights today I am honored to be speaking with Professor Mark SS Professor SS is a neuropsychologist and psychoanalyst best known for his discovery of the full brain mechanisms underlying dreams and his integration of psychoanalytic theories with modern Neuroscience more recently he's been focusing on active inference and what it might bring to bear on the hardest problem of them all Consciousness mark thank you so much for joining me it's a yeah as I said in the introduction it's a real honor and privilege it's it's great to be here Darius thanks very much so yeah I mean you have been in the field of cognitive Neuroscience for many decades but I guess have only recently in the last 10 years or so have really taken an interest in in active inference and seeing you know as many people have I think when I I read your book um The Hidden Spring which is a wonderful read and I think what I really liked about it I've been looking I'm very interested in the hard problem and trying to kind of see different perspectives with which we can take to to approach the hard problem and I've been looking for what I've been calling the synthetic a priori of Consciousness and what I mean by that is can we have a can we determine a function of cognition that needs to be conscious a priori so for example you could say attention well I can imagine offline attention I'm just sort of pinging around without it being online and I thought you got very close and I I will not say determin you know I will not say in a determined fashion because David Charmers will shs at me but I think you got very close with this notion that affect is a good candidate for what I would call the synthetic a prior of Consciousness now affect is can be seen as a kind of loose term so I think where we might want to start is just if you could give a kind of brief definition of what you mean by affect because obviously um maybe what you and yach panp in basic emotion Theory think is a might differ from a constructionist account of aect and so on so maybe if we could just start laying down what affect is before we get to Consciousness thank you um yes that's a good uh point to start with because as you say different people do mean different things by that term um so what I mean by affect is first and not not foremost but first that it is valanced in other words there's a goodness and Badness um to affect and um appropo of your opening remarks about a um aiori synthetic function uh the I think I should mention that the goodness and Badness of affect um is it applies only to the subject of the affect so it's good from my point of view or bad from my point of view it doesn't mean that it is objectively or universally good or bad so an affective state uh is a valanced state it has goodness and Badness the second um and perhaps this is the uh when I said I don't want to say first and foremost veilance because perhaps this is the foremost characteristic is that it is qualified by which I mean uh it has a specific uh differentiated quality there are there are different affects so although all affects share the the um property of veilance of goodness or Badness um each affect has its own unique quality um I don't mean to say that they aren't intermediate forms of affect but what I mean is that the that the feeling of hunger uh is qualitatively different from the feeling of thirst is qualitatively different from the feeling of sleepiness and so on and uh this is important um because you you you you can't you can't say um I have uh currently six out of 10 of sleepiness and six out of 10 of thirst therefore um I have uh 12 out of 20 of total bily need and all I need to be is sleep uh if you if you did that you would reduce the the the the common uh denominator uh the the the continuous variable of need but you would die uh that is because uh sleepiness and thirst are not the same thing they are qualitatively different they are categorical variables so so that's for me uh the essential features of affect that they valanced um which which implies they're subjective their veillance is subjective uh and they are categorical uh qualitatively differentiated uh States okay that's really useful so I think before we get into the qualitative distinctions because I you know uh people may have seen your debates with Lisa Felman Barrett um and how deep that distinction runs was kind of a sticking point in your in your conversations before we go there we don't even necessarily need to go there I think the really interesting point there is that good or Badness is only good or Badness for a subject I've been thinking about this in terms of relevance or prioritization that whatever enters into the space of Consciousness is always prioritized or relevant to the uh to the agent now I think there's a second question that I will come to about are we not putting the C for the horse by H inferring the presence of an agent for whom something is good or bad but before I go there folding in something like salience relevance or priority is that uh built into the goodness or Badness of the signal of the phenomenal state or is it or does it necessarily underly or underwrite that phenomenal State um the way I look on it is that the the underlying category is a category of need uh you do not feel all your needs so for example your need for energy resources um is met by your constantly burning up the uh glucose uh deposited in your ad tissues but you don't feel anything um when that um need is prioritized you then feel hunger um so the prioritization and the becoming conscious of the need are I think intimately bound up with each other and so what therefore does that say for this kind of synthetic aior of Consciousness in so far as if relevance or salience or priority whatever you want to say is it it's difficult to imagine being conscious of something which is not in some way Salient to you does that put the that notion of salience uh which we could you know I've been proposing that Consciousness might be something like a relevance filter or a salience filter would that make that function that relevance function more akin to this synthetic aior of Consciousness than affect per se I think that uh the answer to that is bound up with uh what do we mean when we say we we are prioritizing one need over another it's not it's not the prioritization is not an end in itself it's a prioritization for Action uh you can't you can't do everything uh at once um in terms of um this prioritization uh uh uh imperative and I'll explain what I mean by that uh and has everything to do with when you say an aoi synthetic function um it has everything to do with what function we are talking about so what I'm saying is that that function is not merely the function of need or the registration of a need uh it's the feeling of the need and what functionality is added by the feeling of the need has everything to do with the action bottleneck that I was talking about the prioritized need is felt so that you can monitor how well or badly you're doing uh in your attempts to meet that need the nonprioritized needs are relegated to automaticity the actions that you perform in relation to the nonprioritized needs do not have to be felt because uh they are being met or at least attempts are being made to meet them outside of Consciousness by existing uh fixed um policies the prioritized need which is felt uh the the function of the feeling is precisely because you're in a state of uncertainty um the the the automaticity uh doesn't cut it automaticity is dangerous uh because you don't know what to do so you're feeling your way through a novel situation or through an unpredicted situation and um the the the feeling tells you whether what you're currently doing the policy you're currently following whether it's working or not so it underwrites Choice uh it underwrites the possibility uh of changing your mind on the Fly um and then of course learning from the experience which which which results in the laying down of new predictions which may in future uh be executed unconsciously it's while you it's sort of predictive work in progress I suppose is a is a good way um of describing it and I want to um make explicit what I in something in what I've just said when I said that this underwrites Choice um it means that it underwrites voluntary action so this is the mechanistic difference in my view between voluntary and automatic or autonomic action so yeah okay that's that's interesting there are some things that I think we can fold in from an active inference perspective because um you were talking you were talking about almost metacognitive monitoring of how well you're doing at minimizing free energy and that actually has been fleshed out in in active inference as a COR jofy and Corelli paper all the way back in 2013 but recently people have been talking about um era Dynamics which is this awareness of how well you're doing at minimizing free energy over a span of time and they have Associated that with veilance my question here is more therefore philosophical so it's not really pulling that in aart um although there might be some people who who would encourage me to do do so but rather why couldn't that uh monitoring be done offline so what is it about and this really gets to the heart of this notion of a synthetic priority of Consciousness I can imagine in some sense a robot in some fault experiment who was able to offline attribute goodness or Badness to their experience and use that in a functional manner to guide their expected free energy calculations let's say what is there something critical to experience per se which makes it adaptive um for these free energy minimizing systems and their metacognitive capacities to track their ability to minimize free energy yes I believe that there is before I um explain why I just want to flag maybe for later discussion so I don't I don't divert us from the question that you're the line of questioning that you're currently following that I'm not so sure that I would describe the mechanism that um I just articulated as a metacognitive uh mechanism I I think there's a difference between um the feeling um and what the feeling is about in other words there's a difference between uh the the the raw feeling which is the registering uh of what's happening to your in in the terms that you were just using uh to your free energy and um you're cognizing uh representing to use another language um what uh in in the context within which I'm feeling this what is causing me to feel this so sure I yeah I certainly I certainly agree with that yeah I didn't mean to say in terms of metacognitive purely this notion that it can be done in some pre-reflexive non-conceptual non-representational manner purely this notion that there is at least embedded in phenomenal space some self referential but it could be very low level like lived experience I am doing well I am doing badly but it doesn't have to be a higher order narrative um I doing well I'm doing badly okay good now um to your question you ask why can't this function that I've just described uh as the mechanism of feeling uh you're asking why can't it go on in the dark uh why why why is it why is it necessary for it to be felt so let me if you'll just bear with me for a for a couple of minutes let me unpack the mechanism in in slightly more pedantic detail um we're talking about a self-organizing system uh which is which is intrinsically imbued uh with the um the

the um function that it must continue what it does uh must uh maximize its chances of continuing to exist so that's a sort of pre that's a starting point of of of any complex dynamical self-organizing system so there is a kind of intrinsic I don't say kind of there is an intrinsic intentionality in such a system it's trying to achieve something I don't mean it's metacognitively trying to achieve something I just mean it has it has a goal directedness let me let me put it that way um now it's I then said that um the monitoring of how well or badly I'm doing uh this this can this veilance function um that's what veilance is because it's just to the extent that what I'm doing um is moving me out of my viable bounds is making me less likely to survive to and vice versa you know when it's moving me back towards my viable bonds that just is good or bad for the system the but again I need to emphasize the point that I emphasized earlier it just is good or bad for the system in in in uh Nagel's sense it's it's it's good for the system and good to the system you know so it's not it's not good or bad in any objective sense it's good or bad only in a subjective sense now you're saying why can't that sort of monitoring uh be done um in the dark I'm saying it can that's why I made the point earlier about the difference between the continuous variable of total need or total free energy um and the categorical variable um of which which need am I talking about here categorical variables just are distinguished qualitatively as I said earlier thirst and sleepiness just are not the same thing uh so they are categorical variables uh that means they're qualitatively different so what I've just described is something which mechanistically is a felt state I'm saying I'm saying from the point of view of the system there is an an existential uh variable uh called veilance in other words this this is uh good or bad for me uh for my continued existence whether or not this is metacognitively reflected it doesn't matter there is this thing which just is good or bad and just is subjective and it just is qualitatively differentiated so if you're saying that there's a certain function U that the system has uh which is intrinsically subjective intrinsically valanced um and intrinsically qualitatively distinguished I'm saying that is a feeling so if you say say how can that go on in the dark I would say what do you mean uh this intrinsically is something that is felt that's the me a mechanistic description of what a feeling is and I think that the crucial step um that uh needs to be um uh identified there um is that is it justifiable to take the point of view of the system in other words what is it that justifies one uh to take the Viewpoint of the system and I hope you can see in what I said the in the intrinsically subjective nature of the things that I was talking about there that justifies us the self of the system justifies Us in taking the subjective point of view of the system and if you do that from a subjective point of view the mechanism I've just described um just is a feeling that's what we mean by a feeling so um I I I I hope I've been plain clear enough you have well you have been for me I mean I'm familiar with these arguments of sort of intrinsic intentionality I mean as you know they go all the way back to haiger and this notion of taking a stance on your own being and the idea that it's kind of open-ended we are naturally as you say directed towards this kind of open-ended Futurity and that's gets fleshed out in you know I've recently just finished Evan Thompson's mind in life and he is wonderfully clear on that about this idea of kind of we make we the unel comes out of our own stance that we take upon the world I guess I still have a kind of residual Charmers skepticism which is does and I think therefore it's necessary to get kind of clear on the axioms that you're taking here does taking an intentional stance on your own being require Consciousness maybe intent well intentional is actually not necessarily the wrong word here but I mean can you have a self-organizing complex dynamical system and I think there's an interesting point here about the imperative but I'll come back to that but can you have a complex self organizing system which is autopoetic in the sense it it is intended towards its own um self persistence which need not be conscious or at least conscious in the way that we think of a conscious as subjectively conscious I can almost imagine that you have a nonsubjective Consciousness I.E an objective Consciousness if one could imagine and such a thing which might in some ways uh suit that self-organizing process because it gets rid of bias and uh overfitting your data and confirmation bias and all this other stuff I recognize we've gone slightly off the beat and track here but I'm just wondering whether there's like one thing that I'm missing maybe it's an axiom or maybe it's an argument I'm not sure about exactly why autopoetic self organization entails subjectivity per se um well well let me start um with an appeal to Authority which is not a good thing uh and then uh let me add a um semantic point which is not a good thing and then I'll I'll answer your question properly sure the appeal to Authority I have to say um I was very pleased that Tom Nagel wrote a commentary on my book in the journal yeah he wrote it in the Journal of Consciousness studies um and there he he said in in in relation to exactly the question you've just asked that he thinks that it's it's a non nonstarter the idea of an affective zombie so he accepts that it cannot be an affective zombie this this is a this is a nonsecretor um but that's an appeal to Authority I'm just saying no less a person the nature of subjectivity uh he he says he he accepts that there cannot be an an affective zombie in other words there cannot be a system equipped with the functionality that I've just described that is a zombie because if it were a zombie it wouldn't have the functionality I've described but that's just an appeal to Authority then there's the semantic Point um I want to make clear that I'm talking about feeling okay and you asked me earlier about affect and you said different people use the word in different ways and they mean different things by it um I Hope from my description of how what I mean by the word you can see why I use it synonymously with feeling um I like to use the word feeling when I'm discussing the issues that you're raising because it just is an oxymoron to speak of a feeling that you don't feel uh there's it's not a feeling if you don't feel it uh and and so the thing denoted by that word um is the thing I'm talking about now I said already that I accept that that's just a semantic point and you know but it it it leads to something more than a semantic point when I say the thing I'm talking about is the thing denoted by the word feeling therefore the thing I'm talking about cannot be unfelt because uh that's the thing I'm talking about the thing I'm talking about is the thing that we call feeling now that leads me to the substantive answer to your question I really do believe that this this question as you formulated it um I I I recognize of course as you formulated it derived from or with um with reference to David chs's formulation of it um I don't think this question would ever have Arisen if it wasn't for the fact that uh we were using as at the time that Charmers wrote his famous paper and the book that followed it um and formulated this term the hard problem of Consciousness um we were using as our model example of Consciousness visual Consciousness uh um This was um perhaps uh inevitable because Consciousness is dominated by Vision uh and uh it was perhaps um because Francis Crick used it as his model example in his book uh the the the um uh what was it called the astonishing hypothesis which came out one year before uh charm's paper um the thing is that I think that was the wrong place to start that all of those questions arise only because vision is not an intrinsically conscious function uh even the highest of visual um functionality uh in other words uniquely human cortical um visual functions like reading with comprehension can be done without Consciousness so the question why doesn't all this information processing go on in the dark without any inner field which is to quote exactly the the phraseology that Charmers used I think that that's a legitimate question and I can see why the puzzlement arises because you can do exactly the same thing without Consciousness so you know like reading with comprehension why on Earth do you ever why does there have to be something it is like to read if the exact same thing can done without there being something it's like to do it now you then once you've got this puzzlement then somebody like you a reader of Charmers comes to my argument and says well why is it inconceivable that this thing couldn't be done by an affective zombie and I'm saying well that's a silly question because look at what I've just told you you know I've told you what the function is I've told you that it is intrinsically subjective it's intrinsically existential it's intrinsically veence uh and it's qualitatively distinguished and if you describe such a mechanism which is necessarily registered by the system then you're describing a feeling so what do you mean it could be not felt it wouldn't be a feeling if it was not felt and I've told you what the feeling does you know and yeah I guess I guess I need to be a little bit more philosophically robuster and it's come to mind here that one can take charma's claims and and that 95 paper and the book in multiple ways so one is why um couldn't this functionality be done in the dark and I take and I agree actually that feeling a prior in and of itself by definition has to be felt so that's something that you can have which needs to be felt so that that notion of in the darkness doesn't really stand when it comes to affect the another the other part of David charma's argument is how do neural processes give r rise to subjective states to what it feels like to be me or you and I guess that is a different direction which I presume you probably don't think there is an answer to and I guess what that leads me back to is when I'm talking about the synthetic a prior of Consciousness sure feeling gets us there I guess what I was saying is if we go if we have the sort of view from nowhere we can go back all the way to the evolutionary start and let's say we did start as non-conscious beings why was it functionally adaptive for human beings to develop feelings in the first place right if this metacognitive non-representational uh self-referential process what it became imbued with feeling not I'm not going to ask how that happened because no one knows but I guess the question here is why did that happen and that's when I say well couldn't couldn't have happened in the dark sure feelings themselves can't happen in the dark per se but the function that feelings fulfill namely this monitoring process why why did that have to evolve so as to contain qualia subjectivity and and so on that's the kind of Crux of the question I guess um I think you're I think either you're asking three questions or I've got three answers uh to your question so I'm so I'm not so okay so I don't okay I think there's actually only one here one okay so I'm not fussed about how I'm gonna lose my train of thought I'm an old go go go go go go please me go go go so um if I haven't already lost it um ah yeah what I want to say first of all is the um Mary uh the visual neuroscientist um if if if she were an affective neuroscientist uh if she were to describe the to know everything there is to know about the mechanism of feeling it would necessarily predict that it feels like something and and she would know something about what it is like to have a feeling from the mechanistic account of it um that's that's that's the the first thing that you were saying in your question brought that answer to my mind but I think you said you concede that point uh I'm not uninterested I'm uninterested I well now I'm interested but I was uninterested in the how and I'm and I'll give you the feelings need to be felt I guess my question is evolutionarily why does the function of feeling need to be felt yes so um that leads to or or that to get to that I've got to go via a second thing that I I heard you saying which is um you know how does the physiological process cause the subjective feeling State and there I have to make clear that I am a dual aspect monist um I'm you may want to explain what that is for the audience yes I I I will um I'll put it uh very simply uh when I wake up in the morning uh before I open my eyes I experience my existence I think to myself I am Mark Psalms thank God I'm still alive that's my mind okay then I stagger over to the bathroom and I look in the mirror um and I see this body uh uh called Mark alms and I say uh there is further evidence that I still exist thank God uh now please note both of those things I'm calling them Mark salms u in other words these are two um two phenomenal uh two appearances two observational perspectives upon one and the same thing so there is the subjective being of Mark PMS that I experience mentally when I wake up before I open my eyes and then there is the objective bodily uh thing called Mark salms that I experience visually uh when I look at him in the mirror that's the dual aspect monus says these are not two different things not two different uh ontological entities there's one underlying ontological entity uh and that's called Mark salms so then the question is which is an abstraction then then the question becomes well what stuff is Mark SS made of and I'm saying Mark SS is made of a functional mechanism uh and which the one I was describing earlier uh is a self-organizing therefore in biological terms a homeostatic system uh homeostasis is a mechanism it's an abstraction there's there's there's we infer the underlying mechanisms uh from the the The observed data uh when it comes to the the apparatus of the Mind the thing that we are talking about here which is also the apparatus of the brain it's that apparatus that we're talking about it's the it's the the functional principles governing this part of nature that we infer from the two observables so when it comes to um for example memory you know you can have a memory that's the experience of a memory or you can see optogenetically the neurons that are uh activated uh whilst you are having that same memory these are two perspectives upon one thing which is an abstraction called the memory system or the function of memory so uh I think that that is the actual thing that we're studying in cognitive Neuroscience um although I don't like to call it cognitive Neuroscience because that seems to exclude affec of Neuroscience so the thing the thing that we're studying in what we're let's call it neuros pychology which is a slightly less prejudiced word um in this in this mind and brain science I think that the the um the the ultimate object of study of neuros pychology is things like the memory system uh the perceptual system U the executive system and so on we're trying to discern the laws uh the underlying monistic laws uh governing this part of nature that we can observe from these two different uh perspectives both both the objective and the subjective one now um sorry if I if I'm getting quite abstract no no no to come back very useful to come back to the essential Point um if that is the framework uh the metaphysical framework within which I'm working and we all have to when it comes to the Mind Body relation uh we have to adopt a metaphysical uh framework and and it's good that we are clear about which one we're adopting that's the one I'm adopting within that framework it doesn't make any sense to say how did the physiological phenomenon cause the psychological phenomenon they are those two phenomena are dual manifestations of a single underlying cause that underlying cause is the mechanism that I spoke of earlier which explains them both um the free energy principle is is is is such a such a thing it it's a it's a it's a a principle which governs uh both how we feel and what brain arousal mechanisms uh we observe physiologically in other words what you see with your eyes and what you feel with your feelings uh are are both manifestations of that single underlying mechanism so to say how does the brain process cause uh the feeling uh is to ask the wrong question uh it's to ask a question that assumes a metaphysical relationship which I think uh is is is the wrong assumption it's like saying if I look um with my eyes I see a flash of lightning um if I listen with my ears uh I hear a clap of Thunder uh and I ask the question how does that visual thing uh the flash of lightning cause that auditory thing the clap of Thunder it's to misunderstand that they're actually both caused uh by a a Geo electrical event uh which we describe abstractly using the the the laws of physics uh using the language of physics or or or or or um ultimate forces you know that that that kind of language it's not something that you observe it's the mechanism that you the thing that you infer that underlies it so um when you're saying you're wanting to know how I'm saying I have answered you how when I describ to you what the mechanism is uh and I very important importantly I need to make this clear the mechanism I describ to you the mechanism of feeling that is um which is which which you can observe from two observational perspectives in other words you can observe brain arousal processes uh and you can observe subjectively feelings uh that these are both manifestations of a single mechanism that's the one that I described to you and this is the important top I said very importantly it also explains why this mean ISM has two um perceptual surfaces why why it has two perceptual perspectives it explains why such a mechanism has a subjective perspective because not everything everything geometrically has a has a subjective perspective in other words you can say there is there is such a thing uh as uh the perspective of being a cup or being a table but that doesn't mean that there is something it is like to be a cup or to be a table so I'm saying the mechanism I've described um which is which is a a mechanism which does not uh exist does not explain the behavior of Cups and tables um is why there is nothing it is like to be a cup or a table so the the the remember that the the underlying manistic ontological entity that we are studying is that mechanism which which which we can study both objectively and subjectively and I've described why this particular mechanism has a subjective aspect now to come to what I understood to be the third part of or third aspect of your question um which is a sort of an evolutionary question you're saying uh why did this happen What functionality is added uh why what is gained by this that that that couldn't go on in the dark I've already explained why I believe that mechanism can't go on in the dark now I want to explain what what it adds um it is in fact I have in a way already explained it but let me articulate it in a slightly different way um a organism let's say an organism because we're talking about living things here but there's another question as to whether they necessarily have to be living things but we could say a system but let's just say a biological system an organism um which has needs uh in other words it has its own um biologically viable States and it has to remain within those States in other words it's a an organism which functions homeostatically um it it it needs predictions as to how to satisfy those needs in other words how to how to return to homeostasis when it's deviating from it and we are born with such predictions us living things have innate predictions uh those are called reflexes those precisely are our autonomic functions so um when I um I I have a I have a a constant um need uh to balance my blood gases there's a there's a balance between oxygen and carbon dioxide that's viable uh for a human being and I have a very simple way of meeting that need which is that I breathe automatically um my my uh intercostal muscles expand expand my diaphragm that sucks in oxygen uh and then uh the the the there's the relaxing of that of those muscles and that expels carbon dioxide and there is actually a need detector in your hypothalamus if you want to describe this mechanism anatomically which is constantly monitoring oxygen carbon dioxide oxygen carbon dioxide but there's one prediction as to how to how to um how to maintain the viable balance and that is this autonomic breathing mechanism now why do we ever become conscious of our need to breathe I know you are now becoming conscious of it because I'm talking about it normally you're not conscious of it um and other words what is added by becoming conscious of it so imagine an an organism equipped with the with the need I've just described uh and with the reflex of prediction the innate prediction the fixed innate instinctual prediction as to how to meet that need that's fine until you place such an organism in an unpredict environment in other words something not predicted by natural selection in other words not the default environment for which this prediction um is an appropriate one the one that you should just breathe at a regular rate in other words for example what happens if you're in a carbon dioxide Filled Room uh you're no longer in the usual uh default environment you're you're in a burning building um and the room is full of carbon dioxide at that point you become conscious of your need need to breathe uh this is the why question that you're asking why uh do you become conscious of your need to breathe it's because now you're in a state of uncertainty as to what to do uh your reflex autonomic uh prediction doesn't cut it uh so you behave stastically you you're in a state of distress uh you you're you you're suffering um Suffocation alarm um or or or respiratory distress or air hunger it goes by various names and why it comes to Consciousness as a feeling is because as you randomly go upstairs uh trying to uh do something other than your instinctual your normal reflex of action it feels worse uh and so that was the wrong decision uh it was a bad choice to go upstairs and this and and the organism registers it in the form of this feeling of Suffocation is getting worse so you change your mind uh and you go downstairs instead and there you get the enormous relief of feeling better you know being able to being able to breathe the oxygen so that that feeling um has guided your voluntary action in fact has has has made possible what we call voluntary action uh it's not just random Behavior because the the your your existing prediction doesn't cut it uh what you do is as I said earlier predictive work in progress in other words you actually feel your way through the problem uh and and and come up with a new prediction which you then lay down that is the Adaptive advantage that is what is evolutionarily um added by this extended form of homeostasis uh in which you don't just have some sort of Auto automatic uh regulator of um that that that keeps you within your viable bonds you have in addition to those automatic Regulators you have a further mechanism uh which kicks in when the automatic regulator isn't working because you're not in the environment that it it expects um and it it it enables you to um to navigate that uh that novel unpredicted uncertain environment I it's one of the what I've just said uh one of the reasons why I I say that that that Consciousness is felt uncertainty that's that's a a py way of describing what it actually is but I hope that I've answered your question there by saying what it adds it adds it adds the the the ability to survive in unpredicted novel environments prior to the um evolution of this um mechanism this extended form of homeostasis this this subjective form of Home extension of homeostasis the what would have happened um is that if you have a population of the organisms that um the hypothetical organisms I'm talking about they all uh have this stereotyped fixed prediction as to what to do if they find themselves in a novel situation because of the innate variation in the genome some of them will do the right thing and and some of them will do the wrong thing in their in their random sort of thrashing about uh the few that do the right thing uh will then pass on uh that uh adaptation uh to the Next Generation in other words the Next Generation when they find themselves in this what was previously a novel situation that will not be a novel situation for the Next Generation the the functionality that I've described supersedes that that is the mechanism of natural selection whereby over Generations um the predictions are improved by natural selection uh the the the functionality I've disc described enables the organism within its own lifetime uh to improve its predictive model um in in um in radically uncertain circumstances and thereby to survive and reproduce there's a lot there um so I want to start by being philosophically skeptical and therefore annoying and say that I still think that this is rest on an axiom that for example Consciousness is there for unpredictability because I still can't see why in principle natural selection couldn't have endowed us with just a very good monitor of what to do in uncertain circumstances so I for I can't imagine why a zombie a philosophical zombie wouldn't be able to recognize as I go up the stairs there's more smoke it's getting worse oh let me go downstairs right and this is just you can just keep fleshing out and having recursive iterations on the mechanism before it needs to come on and become online that is and so the way that I thought about it just to sort of put my cards on the table because I've been acting cryptic is I wonder whether um I wonder whether Consciousness acts as a coar grained representation that allows for Speed and manageable data compress and and and in and genders manageable data compression such that we cannot we can't in principle attend to every single data um sources out there in the world but what Consciousness allows us to do is to compress the data that is in the world into something one that's prioritized and two that can be acted upon in a fast manner whether this is in predictable or unpredictable circumstances and often it will need to be done in unpredictable circumstances but the pure notion that Consciousness is unpredictable is is there for unpredictable circumstances I don't think can skirt the problem or the contention that hypothetic we can never run the counterfactual but there is a there is a hypothetical counterfactual where all of this is being done offline you just have a very adaptive sensor in the hypothalamus let's say who doesn't who who doesn't need an online self model to recognize any signal in some phenomenal way so that's that again I don't feel we to go down this Rabbit Hole um well well because you have articulated your view I will just say something very briefly about it which is that that is in in in in essence and I'm I'm sorry to be so reductionist about you the subtleties in what you've just said but it is in essence a attentional mechanism that you've described it is and uh I'm saying that's fine I spoke earlier about prioritization so that's that's built into what I'm saying uh but uh it's not just any old attentional mechanism it's attention to my homeostatic state in other words to my deviation from my viable bounds it's not attention to anything in the world in other words just unpredictability I'm saying no it's not just unpredictability it's unpredictability as to how to satisfy this need in other words it's a certain kind of unpredictability for a certain kind of system I think that that is the ground zero of feeling that is how it comes into existence so it is by the mechanism you describe but only applied to a particular part of Nature and not not to everything in the universe that's a that's a very useful ad I've actually just written this paper uh with the help of Carl and L Costa about sort of eliding the endogenous exogenous attention distinction within active inference by saying that attention is basically as as we've been speaking about a priority filter but always for the sense making all organism um and and and in active inference you can say well and then you flesh that out and say well the deepest most precise preferences are encoded at the deepest level of the generative hierarchy hence why I don't care where my phone is when I'm suffocating for example because the preference I have my phone is far less precise and far higher than um that is exactly how I see it yep yeah okay so so I wanted to ask you said something about so we have dual aspect monism that we can describe to phenomena we can take two perspectives on a phenomena like memory for example we can think of the hippocampus and we can also think of what it is like to have a memory and you said that what the real Target of cognitive Neuroscience or neuros pychology should be is the single mechanism that underlies both of those perspectives and you preferred the active inference or the free energy principle might be one of those mechanisms I wonder whether this was relating to what people talk about in ter in Act of inference as a dual information geometry and what that is is for the audience and I spoke to Van Visa about this actually yesterday so the podcast should be out before this one so people can go there but the idea is that we have an intrinsic geometry which looks at how the actual internal states of the markof blanket evolve over time and then we have the E extrinsic geometry which is what how they are uh manifesting variational beliefs about the external States and one is kind one you can kind of view as if you wanted to take a metaphysical stance or a phenomenal stance one is the view from within and one is the view from without I wonder whether that Jewel aspect fits quite nicely onto your jeel aspect monism or whether you thought about that yes um what you're saying intuitively appeals to me uh when you say I thought about it now for precisely 10 seconds and it intuitively appeals to me there's a paper called um movian monism which is by Van visa and car so that that's where this is yes um I I just wonder you know I I spoke to Van about this yesterday and I said what were the metaphysical consequences before we move on May I just say there's something about uh the there's something about because now I'm remembering that paper um and and I'm remembering what I thought when I read it I think that there is something special about um precision optimization in relation to what you're talking about um in that the Precision um assigned to uh the incoming error signal um in relation to the outgoing prediction um is assed by the system uh to those to those um to that message passing as it were so I think that um there's there's a special place um for precision optimization uh in in in the story that you just described there yeah absolutely yeah I mean in terms of the paper that I just wrote that's exactly it in terms of a more like in terms of a technical Bas graft pomdp schema what we're saying here is sort of higher order preferences govern Precision weighting over What's called the likelihood Matrix the a matrix and that's kind of been you know Felman and Frist and then later papers have proposed that that really is what attention is it's optimizing that Precision waiting yes we not yeah no go only to add that I think a really critical point that I really wanted to stress in that paper is that that's not just Precision per se so Precision I take to be a very strict mathematical construct in uh inverse entropy or inverse covariance whatever you want to say but not all signals which are given a gain you know extra gain synaptic gain or given attention are only those which are precise I.E which have a clear cause rather we we govern our attention is governed fundamentally by our goals even if that the object of attention isn't necessarily precise right it can be very ambiguous but it's very important I pay attention to whether my garden hose is a garden hose or a snake and that will also be affected for whether I'm in London or whether I'm in Sydney so just to add that I think people it's really important to distinguish attention precision and precision waiting as free individual constructs yes that makes good sense to me um so I uh I I I'll put it in the in in more sort of everyday terms that that um uh it has to do because remember I was saying it's a particular type of system and that's another way of saying what you're saying that um if things with respect to the Precision over my policies if things are turning out as expected that's good if uncertainty prevails that's bad and uh so you know this is why it is good to have confidence in your in your policies because it will result in good feelings is absolutely go ahead no but I gathered that you wanted to move on to to another topic well yeah I mean actually just in reference I I like making sure that people know I kind of feel like I should let people know where they can go to read all this stuff so just in terms of the Precision over your policies this is known as the gamma parameter in active inference and if people want to read about that Casper hesp 21 2021 paper is a really good place to start um where he talks about veilance in terms of this gamma Precision okay yes I so I um I watched your two conversations with Lisa Felman Barrett and greatly enjoyed them uh and and yeah I thought they were really fascinating and the Very novel way of doing scientific discussion because normally it's very angry and on Twitter um I thought a really interesting thing that came up although it was only really briefly touched upon was kind of the distinction between interception and extra reception and I I found actually I've never really thought that it could potentially be quite a blurry line but I feel like Lisa was Keen to say that some of the things that you were proposing were fundamentally extra receptive I a pin prick or something like that could also be construed as inter receptive so if we say okay let's take the pin prick example and let's talk about aect so you're not I if I'm say if I'm correct you wouldn't believe that all affect is fundamentally interceptive is that a fair play fair thing to say oh yes I absolutely disagree with that view but slightly comically in relation to the discussion with Lisa felin Barrett that you're referring to that was the starting point of that discussion was that I said I disagree with her and Neil Seth uh in their in their associating affect with interception for them interception is affect and interception is this other form of perception remember to them aect is just a modality of perception it's an interceptive modality of perception and I was saying no I don't agree with that at all I don't think that it is and then I used the example of pain and I said what could be more extra receptive you know than this thing is from the outside being stuck into my finger um or a fright you know there's something happens out there in the world and that thing out there in the world startles me so it's exteroceptive and so I was saying affect uh clearly is not extra receptive then she got into these gymnastics that you're referring to of saying well you know if you do this then a pinprick is interceptive and I say well fine you know if you do this then that's fine but to me the crucial thing is that that it's neither the extracephalic the feeling uh is the homeostatic deviation homeostats are not they are not interceptive or EXT receptive they both they are registering a a configuration um which it uh which it infers this uh this is bad for me or this is good for me in relation to the particular category of need that we're talking about so so I don't think you and Lisa would necessarily disagree on that if I listen to that conversation correctly because I think she was keen and I guess it's a fundamental part of her theory that there isn't there isn't necessarily meaning in the data signal itself and it's only in its uh collision with the prediction that we get in basian terms a posterior which relates to our actual fundamental prior and that Divergence is what you might call a so I you wouldn't disagree on that can can I just make clear there because you know these disagreements with Lisa I think are very important uh for the future of our field by which I mean by I mean affective science that that's the field I'm talking about that's that what I share with Lisa is an interest in the in the fundamental nature of affect or what she calls emotion because as you said at the beginning affect and emotion and and and so on these words are used differently by different people for her a does not have quality it has veillance and arousal but it is not distinguish in in its intrinsic um so um the you're quite right in everything you just said there that she and I don't differ in the things that you just said there but for me the crucial point and you can see in relation to everything that I've said to you already why it's crucial uh is that I think that the starting point is that those homeostats are innate you know that we are born with categorically distinct homeostats um and the I won't repeat everything I said earlier but it is from those homeostats uh that the uh ultimately that the that that the quality of feeling arises when I was saying thirst feels different from hunger and so on and remember I'm saying these are not only inceptive the Fright feels different from Pain these are extracephalic side it's extracephalic from this category of need and if you prioritize that deviation in other words this is the one that I'm going to now I'm going to my un this is the most Salient category of need uh this is the one where my I'm going to palpate my uncertainty as I act in the world uh this is in other words the one that I'm going to that is now going to govern my voluntary behavior that the quality of the feeling uh that governs uh that action cycle um is going to arise from the category uh of need that we're talking about in other words from the homeostat her view is that there are no innate categories of emotion and I've just explained why I don't agree with that that I think is the fundamental difference between her and me and that's what I felt was that's what I felt was the fundamental difference and just so I can get clarity for myself is your view that there is a strict mapping at least numerically between the number of homeostats and the number of basic categories so is there a homat for well these are the ANP and basic emotion you know is there a homeostat for uh seeking is there a homeostat for Rage or lust or whatever I do believe that there's a a homeostat for each one of those what he calls basic emotions um but I need to make clear that those what he calls basic emotions um are not the uh sum total of the basic homeostatic need mechanisms uh that are at work in the in the human or or mamalian or for that matter vertebrate organism um the the the um many of the things that we spoke about at the beginning of our conversation things like hunger and thirst and um and um um respiratory feelings and so on these uh are not emotions these are at least they don't feature in Pip's taxonomy of the emotions because for him emotions are not these things he calls them bodily affects so there are many U homeostats not seven there are many many many many many homeostats in the human organism and it's a very telling fact that not all of them give rise to feelings like blood pressure regulation for example there's no feeling of you know how well or badly you're doing in terms of um in terms of um regulating your blood pressure um and and and I'm just using one example there are many such examples there there ones which are transitional which are Norm nor run autonomically and sometimes we feel them and I just gave an example of that when I spoke about respiration that normally it's it's it's handled autonomically but then at under certain circumstances it comes to Consciousness so I'm sorry I'm probably being overly pedantic but the my point is yes I do agree with you that there for for each need there's a homeostat but there more than seven of them many many many more than seven of them and um only some of them ever come to Consciousness and only some of them might be called emotions in my way of using the word emotion but here's the further very important Point um the fact that they are categorical needs doesn't mean that they aren't amalgams of them uh it doesn't mean that they also aren't higher order metacognitive um uh uh um amalgams and elaborations and so on for her for for Lisa that's the starting point the starting point is that there are these learned categories that we then um that we then there are what she calls emotion Concepts which we then apply to um to what we experience perceptually and I'm saying yeah of course there are these learned Concepts um but they start with uh innate Concepts those are those are the um those are the the the homeostats that I'm talking about how we come to regulate through Al estasis and and and and learning um all of these different mechanisms and and orchestrate them all with each other and so on especially us human beings with all of these higher order abstract Concepts that we use to do so um it includes everything that she writes about I think that what she writes about describes something that definitely does happen but I I just I I just question uh the um the assumption that there are no innate categories to begin with it just makes no evolutionary sense and makes no physiological sense it just doesn't map on to what we know about how the brain works um that that's the that's the point I wanted to make yes that was the that was the big sticking point um and there was this kind of back and forth about what evidence is uh which we don't we don't necessarily need to go into but people can watch it and I thought was very interesting yeah well when when if if you do if you dear listener or viewer do watch it I want you to to to notice one thing which I thought was was was especially important um it's that because Lisa's starting point is that this idea that there are particular circuits for particular emotions and so on which which U of course can be paraded and can be caricatured and can be made to look like an idiotic local I Iz aist stupid idea but but when she says there are no such systems there are in no innate brain systems for innate emotions and

and and other uh categories of organism organismic me when she says that please note in this conversation with me uh I leared that she says the same of um Vision as opposed to sematic sensation as opposed to all faction she says the these two are not innate systems so if that's what she means if if her if her uh um uh way of understanding what we mean by an innate system uh her way of of of using that terminology excludes um the usefulness of speaking about a visual system which we all have and an auditory system that we all have and so on I think you need to understand that it's in that way that she's questioning whether there are basic emotional systems in the brain uh if if if it's such a radical claim that she's that she's making that um we should not speak of any innate systems uh of the kind that we do when we speak of a visual auditory or C sensory and so on system if if if if if emotion is being lumped in with that claim then it's not a claim only about emotion it's a whole different way of thinking thinking about what are useful Concepts um and her reasoning for saying there's no innate visual system is that if you're blind your visual system will be um you know will serve different functions not innately visual well of course if you're blind it's going to go in an anomalous is going to organize in an anomalous way but that doesn't mean that it's not useful in biology to speak of the the fact that in the normal state of things there is an an innate visual system but yes I wonder whether I could you know Lisa's not here to State her piece and so people should go and listen and I'd love to have her on um I'm wondering whether I can sort of Steal man and well actually not steal man either yourself or her but also potentially you know offer a middle suggestion which is could we perhaps propose rather than well I wonder whether Lisa would say she might disagree that there are innate Concepts would she disagree that there are innate processes and by that I mean I can take a s of active inference biological active inference stance on this that there are functions and anatomical systems in the body and brain which have priors to self-organized in a certain way and this is the work of Mike Levin and France cookling and this kind of work about moros space and how that fleshes out and so I wonder whether because I personally and I think this is an argument that goes all the way back to Jerry Fodor and the debates around Chomsky and and innate language I can't see how for example fear would even get ever get off the ground with no innate priming for fear right like I agree fear then becomes socially modulated but how would that categorization like I don't think Lisa would deny that for you and I as fully fleshed conscious propositionally using L language users we can think about fear and have it as a concrete object but how that ever gets off the ground in terms of what makes something fear what make something first without some uh something guiding that process whether it's just a singular concept that acts as a template which things get fit fitted into or process uh like let's say an innate inductive in process something like that as way that someone like Fodor might talk about it and arguably where Chomsky has ended up in in merge and things like this uh maybe that would give us the inductive background from which we can slot these emotions into different categories and in many ways I mean this is this is kind of go you know it takes us all the way to something like a connectionist network which is that these connectionist networks are not completely unsupervised in the sense that they do have prior weights um and so yes I do think the radical tabularasa well one I one I guess I'd have to ask her because again I don't want to speak on her behalf but I wonder whether in steel Manning her position she might have space for something like an innate mechanism or innate process yeah well you're right that um our viewers or listeners must go and watch those they they're on um I think they're on YouTube they are discussions I had on her Channel yes um and um you so you must watch it yourself and and form you know your let and let her listen to her speaking for herself and and form your own opinion but um the topic that you've just mentioned did come up um not very uh um not very it wasn't discussed at Great length but it came up periodically in our two our two uh conversations and uh my understanding what she said is yes she certainly agrees that there are these innate mechanisms uh and they are instinctual behaviors or reflexive action U not necessarily only external behaviors but also internal actions of within the body um but um all that's innate is those stereotyped uh actions um she doesn't think or or or believe uh that there is a feeling that goes with them um so my my given everything that I said earlier you can see why I think that some of them have feeling attached to them because that's precisely what makes you know I am you know suffocating as opposed to I am sleepy uh and uh and you you act accordingly I I think the feeling uh attaches to the category of need um because of what feeling does um you can I believe um um well I don't I I don't go into too much detail but let's just say so I believe that the feeling is attached to the innate category um she accepts there are inade categories but doesn't believe that feelings are attached to them but here's the really interesting point she then says so we then learn we make inferences about these categories so the state I am in now which I have been in before I recognize this state this is called fear um and that's when fear arises fear is the concept uh the Learned concept that you apply to the state that you find yourself in so the first time uh that this um uh in my view the first time that this action uh this instinctual action freezing and fleeing occurs um the the the the the feeling that you will have is fear uh her position seems to be the first time that it happens nothing will occur mentally the second time that it happens you will recognize I've been in this situation before I call this fear and now you formed the concept that's her way of saying it but the crucial thing there is there's no place for a feeling in that why do you have a feeling feeling you've just categorized this state as fear um and she says yes indeed it's exactly the same as any other type of perception it's perceptual inference and um to me that leaves qualia entirely out you know it there there's no need so I asked her the question why then do we feel uh and she said I don't go there that's not a scient question yes well again uh people should go and watch and I will email Lisa again I emailed her and and um I think she was very very busy she's a very busy woman I will I will get back in contact so I couldn't so we're running out of times we only about 10 minutes left but I wanted to very briefly touch upon dreams yes um we haven't spoken we haven't done anything to do with the kind of earlier work of yours as I said you know you're very famous for discovering the forbrain mechanisms underlying dreams in I think the late 90s and you wrote in the well it's I think it's the 2000 paper that what you're what you're really drawing into question there is this Association of REM and dreaming so REM rapid eye movement sleep and dreaming were natur well were historically tied together and you show in that paper that really nicely there's this sort of double dissociation that you can have dreams without REM and REM without dreams and you said at the very end of that paper the biological function of dreaming remains unknown this is at least partly attribut attributable to the fact that the function of dreaming and the equally unknown function of REM sleep have been conflated for more than 40 Years of research future studies of these functions should be uncoupled from one another I was curious one have they been uncoupled from one another and two has that given us any perview or insight into the function of dreaming yes um so the double dissoc thank you for raising this it's that was my um I mean I spent many uh happy years uh doing a dream research um and uh I was very fortunate in the discoveries that I made uh in in that area which like most um rarely significant discoveries happen purely by luck so it wasn't something I was looking for so I just want to make sure that that people understand the double dissociation you're talking about it's not only that you can dream outside of REM and you can be in the REM State without dreaming that is a is an interesting fact but it's a that's not a that's I think was always recognized that the correlation between dreaming uh and and REM sleep is a probabilistic one um you are much more likely to be dreaming in rem than outside of REM um but um they they the the thing that I found which was much more interesting I thought is that lesions which destroy the part of the brain that generates REM which is which is a colonic system in the misop pontan mum that people who who who have lost the capacity to have the REM State because that part of the brain is lesioned uh they continue to dream so with no REM they continue to dream and conversely there are patients um who lose the capacity to dream uh due to a brain lesion elsewhere uh they continue to have REM sleep that's the double dissociation we're talking about standard as well yes indeed and that lesion the one that leads to a loss of dreaming actually there are two lesion sites but the interesting one uh is in the misoc cortical misol limic dopamine circuitry so it's a different circuit from the the one the the one that generates REM sleep it's a different neurotransmitter from the one that modulates REM sleep so so dreaming uh as opposed to REM sleep they correlate with each other but they're not the same thing um and the the there's an interesting question as to why they should correlate with each other because they do very highly correlate with each other but you can totally dissociate them from each other in the way that I've just described now um I I said which is the quotation you just read that we don't know what the function of dreaming is partly because we conflated it with REM sleep so the research that we were doing for decades um into what is dreaming for the the the causal research that we did the the variable that we manipulated was REM sleep so you know it's not surprising that we never got to the answer because we were manipulating something that correlates with dreaming but isn't actually the mechanism of dreaming so I said at that point in the paper that you're quoting that now that we can see that they're two separate mechanisms now we can research the mechanism and function of Dreaming by looking at the at at that mechanism not this one um and so uh that was bound to lead to progress um so the the a very interesting you said so has progress been made a lot of progress has been made and in fact um when I said I spent many happy years in a sleep lab I in fact still am doing some research in that area and so there's one study that's still ongoing uh that um that I've got only the I've only completed the pilot study for it but we're now doing the main study but the findings were so dramatic uh in the in the pilot study that I feel able to say a thing or two about them um the but it's not really in the active inference framework so I don't don't worry no no no no no go go go please can be linked and you know Carl and and Hobson uh um Carl friston and Alan Hobson wrote some very interesting papers within the active inference framework about about dreaming um but the the first thing to say is that uh the this misoc cortical misol liic dopamine system which clearly is the the the driving mechan the causal mechanism of dreaming and there's all sorts of every method we've thrown at it you know single cell recordings uh microdialysis um pharmacological probes lesion studies positron emission tomography everything everything points to that system that system that system that's the one that causes the dream state um the the the really interesting thing about it um is that it is it's been called uh by pank who you men mentioned earlier the seeking system you know the the inal behavior that attaches to this system is foraging exploratory interest and curiosity U it's it it it it drives the most sort of intense exploratory foraging behaviors that an animal is capable of and when you look at the dreaming brain for example bosron emission tomography it's really quite astonishing to see this system is lit up like a Christmas tree but you're asleep you know it's like what the hell how can how can this how can this creature be asleep or this person be asleep if if you weren't told that they were asleep and you just looked at that at that um at that brain image you would say this person's um you know Manic and high on cocaine or something you know so that's the first thing that's that I think is really interesting uh is that uh the the the mechanism that drives the dream uh is a is an intensely motivated active State um and so the the the the the first step toward a an understanding of why do we dream uh has to be you have to start there you know why do we have these intense activated motivated States while we're sleeping and secondly why don't we wake up uh when we have them um so um one way of thinking uh is that the dream happens instead of real Behavior you have you have a virtual reality in which you are um in which you are imagining or deluding yourself uh that you are acting on your motivated States why these motivated states have to happen while you're asleep is another question but the fact that they do happen is is is obvious physiologically demonstrably true so uh the the this state is acted on in the virtual hallucinatory reality of the dream rather than in real reality because um to act on them in in reality would be incompatible with the state of sleep so that is a that is I think a a very simple and reasonable uh hypothesis as to why do we dream we dream so that we can stay asleep during this highly motivated state of mind that occurs State of Mind and Body the if if it weren't for the fact that we're paralyzed we would be running around like lunatics so you you don't run around like a lunatic you because you're running around in your dream you you so how do you test that hypothesis I said earlier that there are two lesion sites which lead to sensation of dreaming uh and we've spoken only about the one the other one is damage to the visual uh Visos spatial cortices uh at the back of the brain uh if you can't generate the me the mental image the heteromodal and predominantly visual but not only visual imagery um that is the dream then how can you dream so they're theoretically less interesting those patients they stop dreaming due to occipitoparietal Strokes um but if the prediction that I've just made the hypothesis I've just formulated as to what the function of dreaming is if it were correct then these patients who can't generate the hallucinated reality because they because of those those lesions they should wake up they should have very bad sleep when when this arousal State occurs it should wake them up so that's the prediction that I tested in my pilot study and let me tell you I have never seen worse polysomnograph than in these patient wow yeah so I think we're on to something there that what what dreams do is they enable you to stay asleep by by letting of steam letting of motivational steam in a in an hallucinatory World rather than the real world I'm not saying that's the function but that's a function that's very cool that's very cool people have to keep their eyes out for that for those sets of papers mark this was awesome we covered so much ground uh I'm I'm really really very grateful indeed I always like to ask people just to say at the very end where people can find them if they have you know if people have questions or people want to read your work how can people reach out to you or where should they find your work so uh to contact me you can contact me by email um it's just my name Mark somms at UCT University of Cape Town doac academic. Za which is South Africa so mark. somms u.c. Za I am also for my sins on what used to be called Twitter but it's now called something else that I don't want to say um so there there I'm markor SS at Mark and we'll make sure to put your Google Scholar and website and all this stuff on in the video description again mark thank you so so very much I enjoyed it thank you all right